

27 - Linear Models with Fixed and Random Effects

Marginal and Conditional Distributions for the Random Effects and for the Target.

The Covariance Structure. Estimation and Inference.

Three Case Studies with Real World Data.

Applied Statistics

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1 The Big Picture — What Problem Are We Solving?

- **Motivation:** Standard linear regression assumes all observations are *independent*. But in many real datasets, observations are grouped — students within schools, patients within hospitals, repeated measurements on the same person.
- Observations *within* the same group are more similar to each other than observations from different groups. Ignoring this leads to:
 - Wrong standard errors
 - Wrong p-values
 - False discoveries or missed effects
- **Goal of LMMs:** Handle grouped/hierarchical data properly by combining two types of effects:
 - **Fixed effects** — population-level trends (apply to everyone)
 - **Random effects** — group-level deviations (each group gets its own adjustment)
- **Example:** Studying how study hours affect exam scores across 10 schools. A fixed effect gives the average relationship for all students. A random effect captures that School A is above average, School B is below average, etc.

2 The Core Model Equation

- The general Linear Mixed Model (LMM) for observation j in group i is:

$$y_{ij} = \beta_0 + \beta_1 X_{j1} + \cdots + \beta_p X_{jp} + b_i + \epsilon_{ij}, \quad i = 1, \dots, N, \quad j = 1, \dots, n_i \quad (1)$$

- In matrix form for group i :

$$\underline{y}_i = X_i \underline{\beta} + Z_i \underline{b}_i + \epsilon_i \quad (2)$$

- Each term explained:
 - $\beta_0 + \beta_1 X_{j1} + \dots + \beta_p X_{jp}$ — **Fixed effects**: the general population trend. E.g., “every extra hour of study increases score by 2 points” for everyone.
 - b_i — **Random effect**: how much group i deviates from the population average. E.g., School A is +3 points above average, so $b_A = +3$.
 - ϵ_{ij} — **Residual error**: individual-level noise that the model cannot explain.
 - X_i — fixed effects design matrix for group i
 - Z_i — random effects design matrix for group i
 - $\underline{\beta}$ — vector of fixed effect coefficients
 - $\underline{b}_i$ — vector of random effects for group i
- **Small example**: 3 schools, population average = 70, study effect = +2 per hour:
 - School A: $b_A = +5$. Student studying 3h: $70 + 2(3) + 5 = 81$
 - School B: $b_B = 0$. Student studying 3h: $70 + 2(3) + 0 = 76$
 - School C: $b_C = -5$. Student studying 3h: $70 + 2(3) - 5 = 71$
- Same study hours, 10 point difference just due to school. This is the random effect at work.

3 Fixed vs. Random Effects — What is the Difference?

- **Fixed effect**: An effect you want to directly estimate and interpret. It applies to the entire population. “How much does study time affect scores?” is a fixed effect question.
- **Random effect**: Used when you do not care about the specific groups, but about the *variability* across groups. The groups are assumed to be a random sample from a larger population.
- **Simple rule**:
 - “I want to know the effect of *these specific* groups” → fixed effect
 - “I want to know *how much groups vary*” → random effect

4 Unconditional Distribution of Random Effects (Section 15.9)

- **Question**: Before seeing any data, what do the random effects look like?

- The random effects follow a multivariate normal distribution with zero mean:

$$\underline{b} \sim \mathcal{N}(0, D) \quad (3)$$

- **Why zero mean?** Because b_i is a *deviation*. Some groups are above average, some below, but on average they cancel out to zero. If all groups were above average, we would have estimated the mean wrong.
- The variance-covariance matrix D is written as:

$$D(\sigma^2, \underline{\theta}_D) = \sigma^2 D(\underline{\theta}_D) \quad (4)$$

where σ^2 is the overall noise level and $\underline{\theta}_D$ is a vector of parameters describing the structure of between-group variability.

- **Two structures for D :**
 - **General (Unstructured):** Any two random effects can be correlated. Matrix D must only be symmetric and positive-definite. Has $q(q+1)/2$ free parameters: q variances and $q(q-1)/2$ covariances. Hard to estimate when n or q is small.
 - **Simplified (Diagonal):** All random effects are independent. D is diagonal with q variance parameters and zero covariances. Much easier to estimate.
- **Example:** Random intercept + random slope ($q = 2$). Unstructured D has 3 parameters: $\text{Var}(\text{intercept})$, $\text{Var}(\text{slope})$, $\text{Cov}(\text{intercept}, \text{slope})$. Diagonal D has only 2 parameters.
- **Intuition:** D describes how “diverse” group personalities are. Large diagonal entries mean groups vary a lot. Large off-diagonal entries mean certain traits tend to go together across groups.

5 Conditional Distribution of Random Effects (Section 15.10)

- **Question:** If we *know* the random effect b_i for group i , what does the data in that group look like?
- The conditional mean and variance are:

$$E[\underline{y}_i | \underline{b}_i] = \underline{\mu}_i = X_i \underline{\beta} + Z_i \underline{b}_i, \quad \underline{\mu}_i = (\mu_{i1} \ \cdots \ \mu_{in_i}) \quad (5)$$

$$V[\underline{y}_i | \underline{b}_i] = V[\underline{\epsilon}_i | \underline{b}_i] = \sigma^2 R_i \quad (6)$$

- **Plain language:** Once you know the group’s random effect, the data is centered at the fixed effect prediction plus the random adjustment, with noise $\sigma^2 R_i$.
- R_i describes the within-group residual correlation structure. In the simplest case $R_i = I$ (identity), meaning residuals are independent and identically distributed.

- **Identifiability constraint:** In the most general form, LMMs are not identifiable — different parameters can produce the same data distribution. A constraint is needed:

$$V[\epsilon_{ij} \mid \underline{b}_i] = \sigma^2 R_{ij} = \sigma^2 \lambda^2(\mu_{ij}, \delta; \underline{v}_{ij}) \quad (7)$$

This ensures the model can distinguish between random effect variance and residual variance.

6 Marginal Distribution — The Key Derivation

- **Problem:** In practice, we do *not* observe b_i . So we average over all possible values of b_i by “integrating out” the random effects:

$$f_y(\underline{y}_i) = \int f_{y|b}(\underline{y}_i \mid b_i) f_b(b_i) db_i = \int f_{y,b}(\underline{y}_i, b_i) db_i \quad (8)$$

- **Beautiful result:** Since both $f_{y|b}$ and f_b are multivariate normals, the marginal distribution of $\underline{y}_i$ is also multivariate normal. The marginal moments are:

$$E[\underline{y}_i] = X_i \underline{\beta} \quad (9)$$

$$V[\underline{y}_i] = \sigma^2 [Z_i D Z_i^T + R_i] = \sigma^2 [Z_i D(\underline{\theta}_D) Z_i^T + R_i(\underline{\theta}_R, v_i)] \quad (10)$$

- **Two components of marginal variance:**
 - $Z_i D Z_i^T$ — **Between-group variance:** how much groups differ. Large if groups are very different.
 - R_i — **Within-group variance:** individual-level noise within a group.
- **Key insight:** Even though observations within a group are *conditionally* independent (given b_i), they are *marginally* correlated — because they share the same random effect. Standard regression misses this.
- **Analogy:** Two siblings. If you know the family genetics (the random effect), each sibling’s height is independent noise around the family tendency. But without knowing the family, two siblings are correlated because they share the same genetic background.

7 The Covariance Structure

- The total marginal covariance is $V_i = \sigma^2 (Z_i D Z_i^T + R_i)$. Its structure depends on choices for both D and R_i .
- **Choices for R_i (within-group residual structure):**
 - **Independent & homoscedastic:** $R_i = I_{n_i}$. Residuals are independent with equal variance. Simplest assumption.
 - **AR(1) — Autoregressive:** For longitudinal/time series data. Residuals closer in time are more correlated:

$$\text{Corr}(\epsilon_{ij}, \epsilon_{ik}) = \rho^{|j-k|} \quad (11)$$

One step apart: correlation ρ . Two steps: ρ^2 . Etc. Used for repeated measures over time.

- **Compound Symmetry:** All pairs of residuals within a group have the same correlation. Used in repeated-measures ANOVA.
- **Heteroscedastic:** Different observations have different variances. Used when measurement noise changes over time or condition.
- **Why does the covariance structure matter?**
 - Wrong structure $\Rightarrow$ wrong standard errors $\Rightarrow$ wrong p-values
 - Use AIC or BIC to compare different covariance structures and select the best one

8 Estimation (Section 15.11)

- **Three things to estimate:**
 - **Fixed effects** $\hat{\beta}$: the population-level trend. “How much does study time affect scores?”
 - **Random effects** $\hat{b}_i$: each group’s deviation. “How far above/below average is School A?”
 - **Variance parameters** $(\hat{\sigma}^2, \hat{\theta}_D, \hat{\theta}_R)$: noise level and between/within-group variability.
- **Remark — Negative variance estimates:** Variance components are non-negative by definition. However, some methods may produce negative estimates when the true variance is close to 0. This happens when the number of groups (q) is small or observations per group (n_i) is small. Modern methods (REML) constrain estimates to be non-negative.
- **Estimation methods:**
 - **Maximum Likelihood (ML):** Maximize the marginal log-likelihood:

$$\ell(\underline{\beta}, \sigma^2, \underline{\theta}) = \sum_{i=1}^N \log f_y(\underline{y}_i; \underline{\beta}, \sigma^2, \underline{\theta}) \quad (12)$$

ML tends to *underestimate* variance components because it does not account for degrees of freedom used to estimate $\underline{\beta}$.

- **Restricted Maximum Likelihood (REML):** Adjusts for the degrees of freedom consumed by fixed effects. Default in most software (R’s `lmer`, Python’s `statsmodels`). Analogous to dividing by $n - 1$ instead of n in sample variance.
- **Best Linear Unbiased Predictor (BLUP)** for random effects:

$$\hat{b}_i = DZ_i^T V_i^{-1} (\underline{y}_i - X_i \hat{\beta}) \quad (13)$$

- **Shrinkage property of BLUP:** Each group’s estimate is pulled toward zero (the population mean). Groups with more data trust their own data more; groups with little data are pulled more strongly toward the population average. This “borrowing of strength” across groups is one of the key advantages of mixed models.

- **Example:** School A has 200 students, School B has 5. Even if School B's 5 students all scored very high, the model says "5 observations are unreliable, pull this estimate toward the mean."

9 Proportion of Variance due to Random Effect (PVRE)

- Also called the **Intraclass Correlation Coefficient (ICC)**.
- Answers: *What fraction of total variability comes from group differences?*

$$\text{PVRE} = \frac{\hat{\sigma}_b^2}{\hat{\sigma}_b^2 + \hat{\sigma}_\epsilon^2} \quad (14)$$

- **Interpretation:**
 - $\text{PVRE} = 0.6 \Rightarrow 60\%$ of variability is between groups, 40% within groups. Groups matter a lot.
 - $\text{PVRE} \approx 0 \Rightarrow$ Almost no group-level variability. A simple regression without random effects may suffice.
- **Policy example:** If $\text{PVRE} = 0.3$ for student exam scores, 30% of variation is between schools. This tells policymakers: invest in schools (between-group), not just individual support (within-group).
- **Profiled confidence intervals:** Focus on one parameter at a time. For random effects, test whether σ_b^2 is significantly different from 0 — i.e., whether the grouping structure matters at all.
- **Scenario analysis:** Use $\hat{\sigma}_b^2$ to quantify practical impact. A group in the $-2\hat{\sigma}_b$ tail vs. the $+2\hat{\sigma}_b$ tail shows the realistic range of group effects.
- **Clustering random effects:** Cluster the estimated $\hat{b}_i$ values to find communities of similar groups. Use group *medoids* (most representative members of each cluster) to summarize each cluster's behavior.

10 Inference

- **For fixed effects $\underline{\beta}$:** Use t-tests or F-tests. The challenge: denominator degrees of freedom are not obvious in mixed models. Use Satterthwaite or Kenward-Roger approximations (implemented in R's `lmerTest`).
- **For variance components σ_b^2 :** Testing $\sigma_b^2 = 0$ is non-standard because zero is on the *boundary* of the parameter space. The likelihood ratio test gives a conservative p-value; the null distribution is a mixture of chi-squared distributions. In practice, use AIC/BIC comparison or profiled confidence intervals.
- **For random effect predictions:** Prediction intervals account for two sources of uncertainty:
 - Uncertainty in the variance parameters
 - Conditional variance of $\hat{b}_i$ given the data

11 Three Case Studies with Real-World Data (Added)

- **Note:** The lecture title explicitly mentions three case studies. While specific datasets are not provided in the notes, the typical cases in Applied Statistics courses are:
- **Case Study 1 — Longitudinal / Repeated Measures:**
 - Example: Blood pressure measured weekly for 12 weeks across 100 patients.
 - Grouping: Patient is the random effect.
 - Random intercept: some patients have naturally higher baseline blood pressure.
 - Random slope: some patients respond faster to treatment.
 - Covariance structure for R_i : AR(1), since consecutive measurements are more correlated.
- **Case Study 2 — Hierarchical / Nested Data:**
 - Example: Student scores nested within schools; workers nested within companies.
 - PVRE tells you how much variability comes from schools vs. from individual students.
 - A high PVRE suggests school-level interventions are more impactful.
- **Case Study 3 — Crossed Random Effects:**
 - Example: Products rated by consumers — both product and consumer are random effects, and they are not nested (each consumer rates multiple products).
 - Model extends to multiple random effect terms:

$$y_{ijk} = X_{ijk}\beta + b_i^{(\text{consumer})} + b_j^{(\text{product})} + \epsilon_{ijk} \quad (15)$$
 - Allows separating consumer-level variability from product-level variability.
- **General workflow for all case studies:**
 1. Explore data and visualize grouping structure
 2. Fit the LMM (choose fixed and random effects)
 3. Select covariance structure using AIC/BIC
 4. Interpret fixed effects (population trends)
 5. Compute PVRE (how much do groups matter?)
 6. Examine random effect predictions and cluster them

TÜRKÇE AÇIKLAMA — Basit Anlatım

T1. Ne Yapmaya Çalışıyoruz?

- Normal regresyon, tüm gözlemlerin birbirinden bağımsız olduğunu varsayar. Ama gerçek veride gruplar var: öğrenciler okullarda, hastalar hastanelerde, ölçümler aynı kişi üzerinde tekrar ediliyor.
- Aynı gruptaki gözlemler birbirine benziyor. Bunu görmezden gelmek: standart hataları, p-değerlerini ve sonuçları bozar.
- **LMM'in amacı:** Gruplama yapısını doğru şekilde modellemek.
 - **Sabit etkiler:** Tüm popülasyon için geçerli genel eğilim.
 - **Rastgele etkiler:** Her grubun genel eğilimden ne kadar saptığı.

T2. Temel Denklem

- i . gruptaki j . gözlem için model:

$$y_{ij} = \underbrace{\beta_0 + \beta_1 X_{j1} + \dots + \beta_p X_{jp}}_{\text{sabit etkiler (herkes için)}} + \underbrace{b_i}_{\text{grubun sapması}} + \underbrace{\epsilon_{ij}}_{\text{bireysel gürültü}} \quad (16)$$

- **Küçük örnek:** 3 okul, genel ortalama not = 70, her 1 saat çalışma = +2 puan.
 - Okul A: $b_A = +5 \Rightarrow$ 3 saat çalışan öğrenci: $70 + 6 + 5 = 81$
 - Okul B: $b_B = 0 \Rightarrow$ 3 saat çalışan öğrenci: $70 + 6 + 0 = 76$
 - Okul C: $b_C = -5 \Rightarrow$ 3 saat çalışan öğrenci: $70 + 6 - 5 = 71$
- Aynı çalışma saati, ama okul farkı yüzünden 10 puan fark. İşte rastgele etki bu.

T3. Rastgele Etkilerin Dağılımı (Koşulsuz)

- Hiçbir veri görmeden, rastgele etkiler nasıl dağılır?

$$\underline{b} \sim \mathcal{N}(0, D) \quad (17)$$

- **Neden sıfır ortalama?** Çünkü b_i bir sapmadır. Bazı okullar ortalamanın üstünde, bazıları altında — ama ortalamaları sıfır.
- D matrisi gruplar arası değişkenliğin yapısını gösterir.
 - **Genel yapı:** Rastgele etkiler birbiriyle ilişkili olabilir. $q(q+1)/2$ parametre var.
 - **Basit (köşegen) yapı:** Rastgele etkiler bağımsız. Sadece q varyans parametresi var.

T4. Koşullu Dağılım — Grubu Bilirsek

- b_i 'yi bilseydik, o gruptaki verinin ortalaması ve varyansı:

$$E[\underline{y}_i | \underline{b}_i] = X_i \underline{\beta} + Z_i \underline{b}_i \quad (18)$$

$$V[\underline{y}_i | \underline{b}_i] = \sigma^2 R_i \quad (19)$$

- R_i : grup içi hata yapısı. En basit durumda $R_i = I$ (hatalar bağımsız, eşit varyanslı).

T5. Marjinal Dağılım — Asıl Güç Buradan

- Pratikte b_i 'yi gözlemleyemiyoruz. O yüzden onu “dışarı entegre ediyoruz”:

$$f_y(\underline{y}_i) = \int f_{y|b}(\underline{y}_i | b_i) f_b(b_i) db_i \quad (20)$$

- Sonuç yine normal dağılım ve marjinal momentler:

$$E[\underline{y}_i] = X_i \underline{\beta} \quad (21)$$

$$V[\underline{y}_i] = \sigma^2 [Z_i D Z_i^T + R_i] \quad (22)$$

- **Varyansın iki bileşeni:**

- $Z_i D Z_i^T$: gruplar *arası* değişkenlik (okullar birbirinden ne kadar farklı)
- R_i : gruplar *iç*i değişkenlik (aynı okuldaki öğrenciler arasındaki fark)

- **Önemli nokta:** Aynı gruptaki gözlemler koşullu olarak bağımsız ama marjinal olarak ilişkili — aynı b_i 'yi paylaşıyorlar.
- **Analoji:** İki kardeş. Ailenin genetiğini biliyorsan her birinin boyu bağımsız. Bilmiyorsan, iki kardeşin boyu ilişkili — aynı genetiği paylaşıyorlar.

T6. Kovaryans Yapısı

- R_i için farklı yapı seçenekleri:
 - **Bağımsız:** $R_i = I$ — en basit durum.
 - **AR(1):** Zamanda yakın ölçümler daha çok ilişkili:

$$\text{Corr}(\epsilon_{ij}, \epsilon_{ik}) = \rho^{|j-k|} \quad (23)$$

Bugünkü ve yarınki ölçüm çok ilişkili (ρ), bugünkü ve 1 ay sonraki az ilişkili (ρ^{30}).

- **Bileşik simetri:** Grup içindeki tüm çiftler aynı korelasyona sahip.
- Yanlış yapı $\Rightarrow$ yanlış standart hatalar $\Rightarrow$ yanlış p-değerleri. AIC/BIC ile en iyi yapıyı seç.

T7. Tahmin

- **Tahmin edilen 3 şey:**

- $\underline{\hat{\beta}}$: Sabit etkiler — genel popülasyon eğilimi
- $\underline{\hat{b}}_i$: Rastgele etkiler — her grubun sapması
- $\hat{\sigma}^2, \hat{\sigma}_b^2$: Varyans parametreleri

- **Yöntemler:**

- **ML (Maximum Likelihood):** Varyans tahminleri biraz küçük çıkar.
- **REML (Restricted ML):** ML'nin yanlışlığını düzeltir. Yazılımların varsayılanı.

- **BLUP** ile rastgele etki tahmini:

$$\underline{\hat{b}}_i = DZ_i^T V_i^{-1} (\underline{y}_i - X_i \underline{\hat{\beta}}) \quad (24)$$

- **Büzüşme (Shrinkage):** Her grubun tahmini genel ortalamaya doğru çekilir. Verisi az olan gruplar daha çok çekilir.
- **Örnek:** Okul A'da 200, Okul B'de 5 öğrenci var. Okul B'nin 5 öğrencisi çok yüksek puan alsa bile model diyor: “5 kişilik veri yetmez, bu tahmini ortalamaya doğru çek.”
- **Negatif varyans sorunu:** Gerçek varyans sıfıra çok yakınsa, tahmin negatif çıkabilir. REML bunu önlemek için tahmini sıfırda sınırlar.

T8. PVRE — Rastgele Etkinin Payı

- Toplam değişkenliğin yüzde kaç gruplar arasından geliyor?

$$PVRE = \frac{\hat{\sigma}_b^2}{\hat{\sigma}_b^2 + \hat{\sigma}_\epsilon^2} \quad (25)$$

- **Yorumlama:**

- $PVRE = 0.6 \Rightarrow$ Değişkenliğin %60'ı gruplar arası, %40'ı bireysel.
- $PVRE \approx 0 \Rightarrow$ Gruplar neredeyse aynı, rastgele etkiye gerek yok.

- **Politika örneği:** Öğrenci notlarında $PVRE = 0.3$ ise: değişkenliğin %30'u okullar arasından. Okullara yatırım yap, sadece bireysel desteğe değil.
- **Senaryo analizi:** $-2\hat{\sigma}_b$ ile $+2\hat{\sigma}_b$ arasındaki fark, en kötü ve en iyi grup arasındaki tipik farkı gösterir.
- **Kümeleme:** Tahmin edilen $\underline{\hat{b}}_i$ değerlerini kümele. Benzer grupları bul. Her kümenin medoidini (en temsili elemanı) kullan.

T9. Çıkarım (Inference)

- **Sabit etkiler için:** t-testi veya F-testi. Satterthwaite yöntemi serbestlik derecesini yaklaşık hesaplar.
- **Varyans bileşenleri için:** $\sigma_b^2 = 0$ testi standart değil çünkü sıfır parametre uzayının sınırında. Olasılık oranı testi kullanılabilir ama p-değeri muhafazakardır. Pratikte AIC/BIC veya profil güven aralıkları kullan.

T10. Üç Vaka Çalışması (Eklendi)

- **Vaka 1 — Boylamsal / Tekrar Ölçümler:**
 - Örnek: 100 hastada 12 hafta boyunca haftalık kan basıncı ölçümü.
 - Rastgele etki: hasta (bazıları başlangıçta daha yüksek).
 - R_i yapısı: AR(1), çünkü ardışık ölçümler daha çok ilişkili.
- **Vaka 2 — Hiyerarşik / İç İç Veri:**
 - Örnek: Okullardaki öğrenci notları.
 - PVRE: değişkenliğin ne kadarı okul kaynaklı, ne kadarı bireysel.
- **Vaka 3 — Çapraz Rastgele Etkiler:**
 - Örnek: Tüketicilerin ürün puanlaması. Hem tüketici hem ürün rastgele etki.
$$y_{ijk} = X_{ijk}\beta + b_i^{(\text{tüketici})} + b_j^{(\text{ürün})} + \epsilon_{ijk} \quad (26)$$
 - Tüketici kaynaklı değişkenlik ile ürün kaynaklı değişkenlik ayrı ayrı modellenir.
- **Tüm vaka çalışmalarında genel iş akışı:**
 1. Veriyi keşfet ve gruplama yapısını görselleştir
 2. LMM'i kur (sabit ve rastgele etkileri seç)
 3. Kovaryans yapısını AIC/BIC ile seç
 4. Sabit etkileri yorumla (popülasyon eğilimi)
 5. PVRE hesapla (gruplar ne kadar önemli?)
 6. Rastgele etki tahminlerini incele ve kümele